OPTIMIZATION TECHNIQUES

PortfolioPilot

Multi-objective portfolio allocation and rebalancing

A hybrid evolutionary and convex approach to Indian-stock portfolio decisions

RETURN

RISK

COST

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The rebalancing problem

An investor owns six stocks. Which holdings should change?

01

Higher return

Choose allocations with higher estimated annual returns.

02

Lower risk

Limit volatility through portfolio diversification.

03

Lower cost

Account for buying and selling when changing holdings.

Output: a Pareto set of feasible trade-offs

Each portfolio offers a different compromise between return, volatility and one-off trading cost.

Data and investment constraints

30
NSE stocks

8
sectors

1,729
daily price rows

2019–2025 data / 756-row estimation window

Portfolio rules

Fully invested, long-only
At most 10 holdings
At least 3% per held stock

≤ 20% per stock
≤ 40% per sector
Buy / sell costs included

Three objectives

Return

maximize $\mu^T x$

Volatility

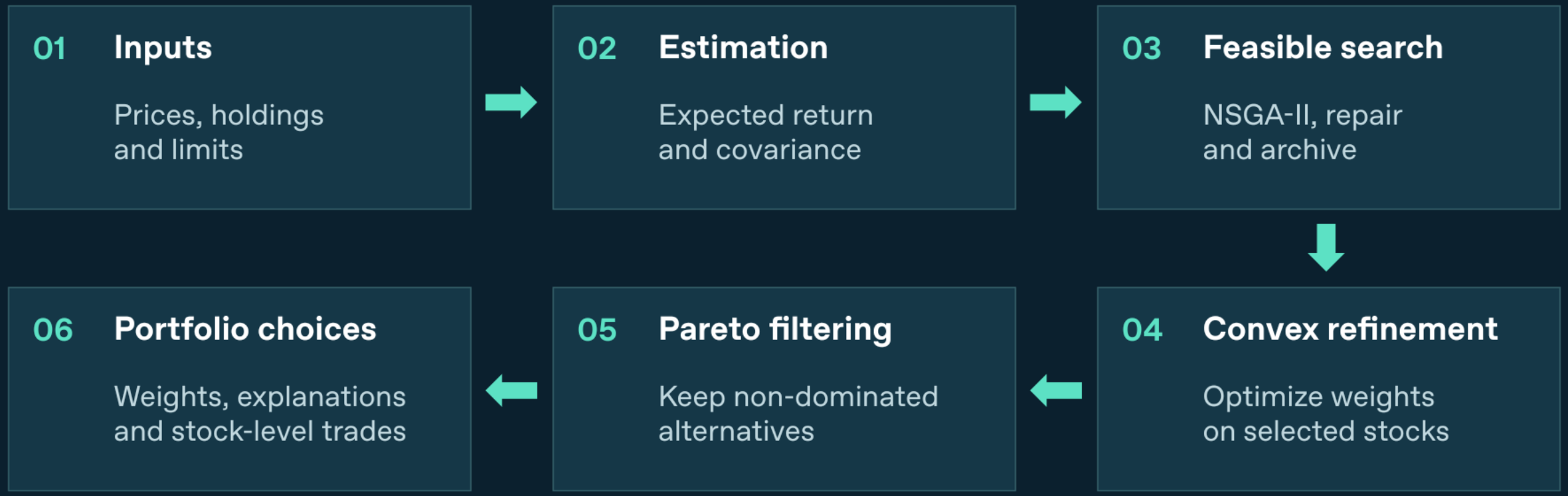
minimize $\sqrt{(x^T \Sigma x)}$

Trading cost

minimize buy + sell fees

System workflow

A feasible portfolio at every stage of the search



Implementation: data loader, constraint model, NSGA-II, convex polish, metrics and frontend engine.

Methods implemented

Method	How it works	Role
Random search	Sample and repair random portfolios	Sampling baseline
Weighted sum	Convex relaxation, support rounding, polish	Solver baseline
NSGA-II	Non-dominated sorting, crowding, evolution	Main search algorithm

30

independent seeds

10,000

candidate evaluations per run

100

population size

Implemented innovations

Evolutionary search + convex polish

Explore which stocks to hold.
Then refine their weights.

01 **Current-portfolio injection**

Preserves the zero-trade-cost option.

02 **Sparse starts + support mutation**

Explores different stock combinations.

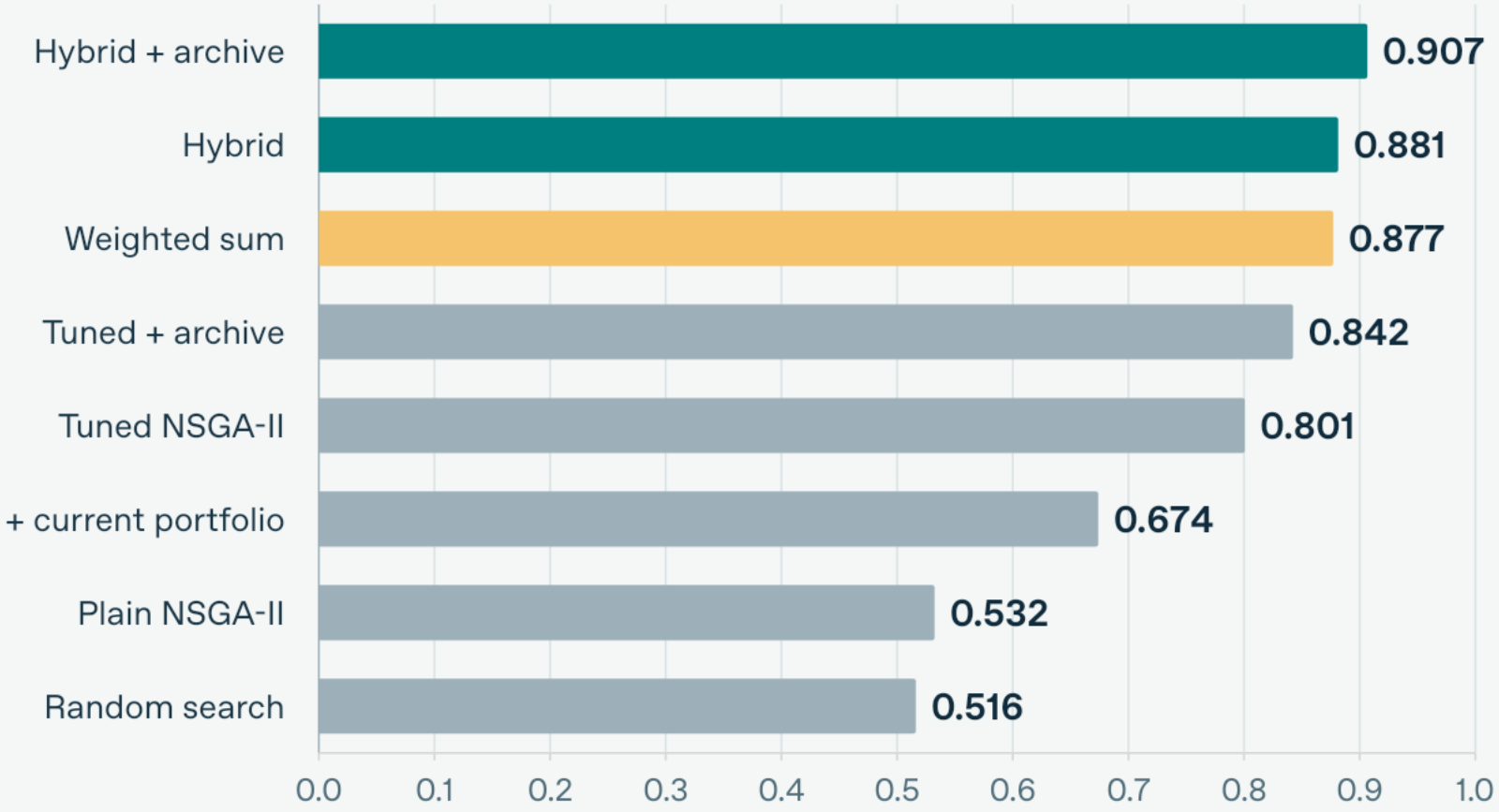
03 **External archive**

Retains up to 200 non-dominated discoveries.

04 **Fixed-support convex polish**

Refines weights with three solver passes.

Hybrid improves Pareto-front quality



+3.37%

mean hypervolume
vs weighted sum

30 / 30

runs beat the baseline

4.68 s vs 0.60 s

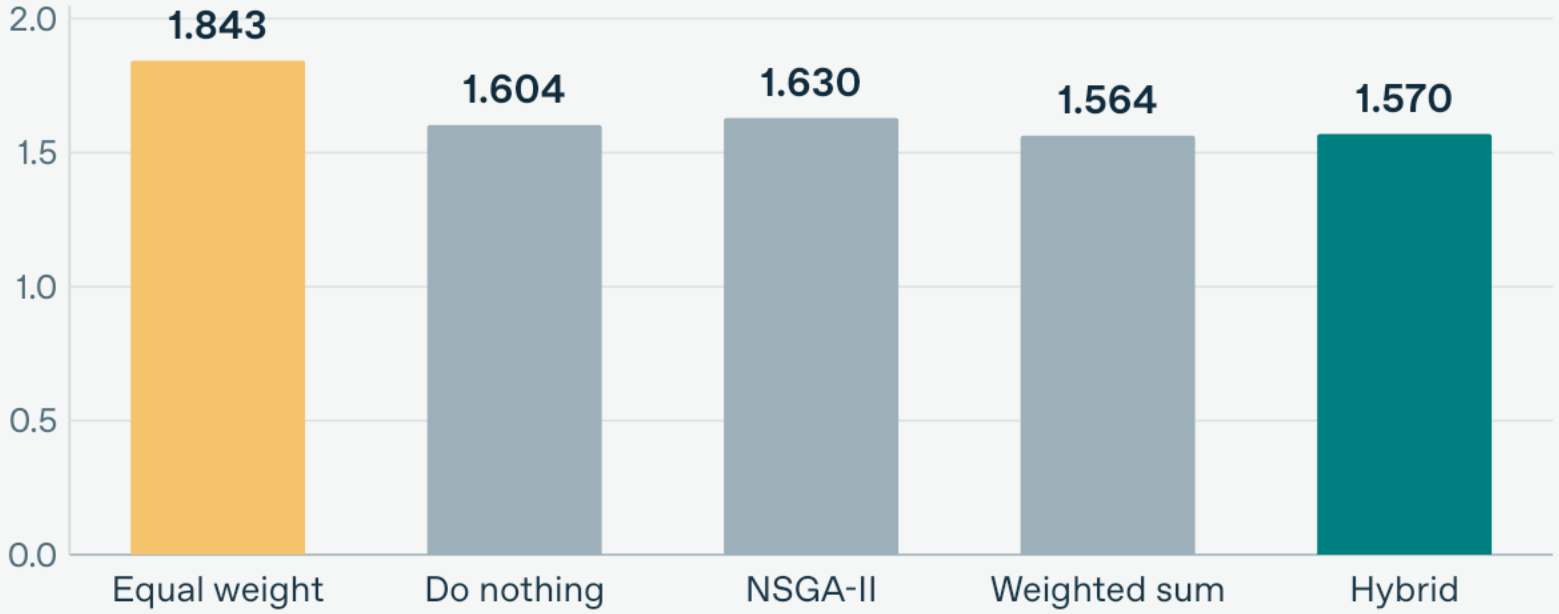
7.74× recorded runtime

Mean normalized hypervolume, higher is better. 30 seeds. Source: results/real/summary.csv.

Backtest: equal weighting finished ahead

15 quarters, rolling estimation, trading costs deducted

Equal weight holds 30 stocks; optimized portfolios allow 10.



Final wealth from unit starting capital

Optimistic return estimates

32.29%

predicted annual return

18.69%

realised quarter return*

Balanced selection rule. *Mean annualized quarter return, not CAGR.

The deployed portfolio comparison



3–6

representative options

Editable holdings and invested capital

Risk–return comparison

Explanations and stock-level trades

Conclusions and Future Scope

Stronger Pareto fronts.

The hybrid improves optimization quality. Historical returns still depend on the quality of the estimates used by the model.

PortfolioPilot

Live demo: portfolio-pilot-deploy.vercel.app

Future work

Shrinkage or factor-based return estimates

Robust models for estimation uncertainty

Stock-specific costs and execution effects

Matched-time and ten-stock benchmarks